

SUPPLEMENT TO CERTIFIED BELLMAN OPERATORS: ALL-RESTART BOUNDS FOR COSTLY POLICY REVISION

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1. FINITE REPRESENTATIONS AND VERIFICATION SEMANTICS

The replication uses a finite rational object $f = (z_i, m_i, b_i, v_i)$: $f(x) = m_i x + b_i$ on each open interval (z_i, z_{i+1}) , while $f(z_i) = v_i$. This distinction is necessary even when the original operating value is continuous. Compiled decisions can jump, a cost-minimizing decision can have a special tie value, and an affine transition can send an entire exceptional point to another exceptional point. Integral-zero sets must therefore remain in a certificate for all restart states.

For a common overlay of two such functions, an affine inequality on an open interval holds if and only if it holds in both endpoint limits. The actual endpoint values are checked separately. This is the reason a midpoint may safely identify which affine formula applies on an already constructed cell, but may not by itself verify that formula's inequality. Every range check uses the limits as well as the stored points. A missing cell, a wrong point value, or a changed model fails verification.

An affine composition $f(\alpha x + \gamma)$ with $\alpha > 0$ adds every preimage of a knot that lies in $[0, 1]$. Values at those preimages are inherited exactly. Maximization first overlays all source partitions; on each common affine cell it inserts all pairwise action intersections. The minimum action index breaks an exact tie. Constrained minimization additionally overlays the binary eligibility masks. Thus eligible singleton actions are not inferred from neighboring open cells.

Normalization removes a knot only if both adjacent affine formulas agree and the stored knot value agrees with that same formula. A separate test creates a point spike, normalizes it, composes it with an affine map, and confirms that the spike and its preimage survive. Exact integration ignores singleton masses only when integrating against the continuous initial distribution. The dynamic recursions never ignore them.

2. CONSTRUCTIVE WITNESS DETAILS

2.1. *Existence and implemented adaptive compression*

For a convex Lipschitz function on $[0, 1]$, a chord interpolant on a mesh of width h lies above the function. A conservative Lh error bound follows from the Lipschitz comparisons to both endpoints. Hence a sufficiently fine rational mesh and sufficiently fine dyadic rounding produce a certified upper approximation at every finite backward step. The maximum over finitely many positive affine compositions of convex piecewise-affine functions remains convex and piecewise affine.

The implementation uses a more economical, checked adaptive rule. Initially retain both endpoints. For each candidate chord, find the source knot with the largest chord error. If that error exceeds half the stage compression allowance, retain that knot and recurse on the two children. This yields an unrounded chord cover with error at most half the allowance. Retained locations are rounded to dyadics, and exact function values at the rounded locations are recomputed. Ordered chord slopes, rather than independent point ordinates, are rounded upward; integrating these rounded slopes from an upward-rounded initial value preserves convexity.

Rounding a retained coordinate can change the chord error. The acceptance decision therefore does not rely only on the error before rounding. It evaluates $G - F$ over the union of all source and encoded knots, including endpoints, and requires $0 \leq G - F \leq \delta$. Precisions 24, 32, 48, 64, 96, and 128 bits for coordinates are available, with 16 extra bits for slope and initial-value rounding. A failed precision ladder is an unresolved construction, not a certificate. All registered cases pass at the first coordinate precision. The implemented cap is 200,000 pieces; the existence theorem does not identify that engineering cap with a universal termination bound.

The original local implementation rounded ordinates and ran timing in a shared Python process. The archived attempt and all its results are retained. The final implementation rounds slopes to preserve the induction invariant and times each neural fit in a fresh process. These corrections do not change the economic model, seeds, training recipe, or tolerance. They are documented in the amendment rather than hidden by overwriting the earlier outputs.

2.2. Upper and lower witnesses

Let $\rho_t = U_t - TU_{t+1}$. Exact compression proves $0 \leq \rho_t \leq \delta$. The upper comparison $V_t \leq U_t$ is independent of the proposal. With $e_t = \eta + \beta e_{t+1}$ and $e_T = 0$, an eligible action satisfies

$$T^a(U_{t+1} - e_{t+1}) = q_t^a - \beta e_{t+1} \geq U_t - e_t.$$

The lower witness is therefore the explicitly constructed function $L_t = U_t - e_t$. It is not a supplied policy-value surface. The stronger realized check uses the exactly evaluated J_t^π and reports $\max_{t,x}(U_t - J_t^\pi)$; this can be considerably smaller than the allowance ε .

Exact installed-policy evaluation is a separate recursion. No raw-policy performance statistic is inferred from final repair. The reference diagnostic $\max(V - v^0)$ uses the uncompressed optimum only after the raw policy is frozen. The deployable zero-intervention certificate instead uses $\max(U - v^0)$ and needs no reference optimum. These quantities are recorded separately as `raw_exact_regret` and `raw_regret_upper`.

3. COST OPTIMALITY, PRESERVATION, AND COMPARISON CLASSES

Given nonempty eligible action sets, the intervention-cost recursion is ordinary finite-horizon dynamic programming with state-dependent constraints. Its output is nevertheless an additional economic object beyond the operating-value solve. The exact checker verifies, at every point where action a is eligible, that D_t does not exceed that action's immediate cost plus its continuation. It also verifies equality for the chosen action. This proves minimum cost and attainment from every restart; a uniform initial density is used only to summarize the resulting functions.

The primary computation does not impose one-step installed-value preservation as an action restriction. Instead, it computes the installed and revised values and verifies the complete difference. All 42 primary outputs preserve operating value. It would be incorrect to claim that this empirical fact establishes preservation for all inputs under the deficit-only action class. The optional intersection with $T^a v_{t+1}^0 \geq v_t^0$, together with a nonemptiness condition, provides that theorem when required.

The additional zero-intervention certificates have a different scope of optimality. Once $U - v^0 \leq \varepsilon$ everywhere is checked, zero is a lower bound on the cost of every feasible policy, not merely policies in the local class. Thus retaining a certified incumbent is globally optimal for intervention cost. This does not prove that a nonzero class minimum is globally optimal for the unconstrained-by-class regret problem. The 12 separate zero certificates explicitly expose this distinction.

The pointwise comparator retains an installed action when eligible and otherwise minimizes current implementation cost with action-index tie breaking. It does not optimize future implementation exposure. Its operating and cost functions are recomputed under the policy it actually deploys. The certified greedy comparator maximizes operating continuation against U . The uncompressed reference maximizes operating value exactly. Neither accuracy-only comparator is mislabeled as solving the constrained minimum-cost problem.

The price envelope is an exact comparison over the listed computed policies. For each pair with different costs, the algorithm includes a nonnegative crossing $(J_i - J_j)/(C_i - C_j)$; it checks all open intervals between crossings and every crossing point. A final unbounded interval covers arbitrarily large prices. Identical lines and exact ties remain valid alternatives. This produces a complete finite comparison, without implying an optimization over uncomputed feasible policies or an empirical monetary calibration.

4. COMPLETE ACTIVE-BOUNDARY PROOF

4.1. *Deterministic wealth and the Dynkin integrand*

Throughout this section the primitives, controls, and restart are those in Theorem 5 of the article. With $p = 0$, $c = 3/4$, and $x(0) = 5/4$,

$$x(t) = \frac{75}{2} - \frac{145}{4}e^{t/50}. \quad (1)$$

For $0 \leq t \leq 1$, $e^{t/50} \leq (1 - 1/50)^{-1} = 50/49$, hence $x(t) \geq 25/49 > 1/2$. It is decreasing and starts below two, so no wealth boundary is reached. Let y denote preference. Direct differentiation of the unchanged settlement gives

$$\begin{aligned} h_\theta(t, y) &= \frac{(3/4)^{1-y}}{1-y} - \theta^2 + 8 - .04\theta(y-2) - .00005 + .002 - \frac{.075}{x(t)} \\ &\quad + .0008(y-2)^2 - .004 \log x(t) + .32(1-t), \end{aligned} \quad (2)$$

$$\partial_\theta h_\theta = -2\theta - .04(y-2). \quad (3)$$

For $y \in [6/5, 14/5]$, put $z = y - 1 \in [1/5, 9/5]$ and $L = \log(4/3) < 1/3$. The utility is $-e^{Lz}/z$, which is increasing on this interval because $Lz < 1$. Its lower bound is $-5e^{1/15} \geq -75/14$. Also $|y - 2| \leq 4/5$, $x(t) \geq 1/2$, and $\log x(t) \leq \log(5/4) < 1/4$. Dropping nonnegative terms yields

$$h_\theta \geq 8 - \frac{1}{20000} - \frac{3}{20} - \frac{1}{1000} - \frac{4}{625} - \frac{1}{25} - \frac{75}{14} = \frac{342357}{140000} > 2. \quad (4)$$

The utility derivative is $e^{Lz}(1 - Lz)/z^2 \geq 10/81$. Thus

$$\partial_y h_\theta \geq \frac{10}{81} - \frac{4}{3125} - \frac{1}{125} = \frac{28901}{253125} > 0. \quad (5)$$

Extend $h_\theta(t, y)$ constantly from $b = 14/5$ to larger y . The extension remains bounded, nondecreasing in y , and at least two. Its parameter derivative retains the bounds in (3). Conservative absolute bounds $|h| \leq 20$ and $|\partial_\theta h| \leq 2$ hold for this extension.

4.2. A coupling lower bound for the derivative

Let $Y_t = u_0 + \theta t + \sigma W_t$, killed at $a = 6/5$, where $u_0 = 61/50$ and $\sigma = 1/20$. If drift increases, the coupled process is higher at every date and lower-boundary survival events are nested. For a fixed nondecreasing integrand $h \geq 2$, paths surviving under both drifts have a nonnegative payoff change, while newly surviving paths contribute at least two. Taking a positive difference quotient gives

$$\partial_\theta \mathbb{E}_\theta[h(t, Y_t) \mathbf{1}\{t < \tau_a\}] \geq 2\partial_\theta \Pr_\theta(t < \tau_a). \quad (6)$$

The direct dependence of h_θ on θ must be added, not included in this frozen-integrand comparison. It contributes at least $-4/125$ for $\theta \in [-1/5, 0]$ and at least $-54/125$ for $\theta \in [0, 1/5]$, after integrating over a discounted survival time of at most one. Gaussian likelihood scores justify differentiability and dominated passage to the occupation integral for this bounded extension.

The reflection formula with drift, differentiated exactly, yields

$$S'_\theta(t) = 16e^{-16\theta} \Phi\left(\frac{\theta t - 1/50}{(1/20)\sqrt{t}}\right). \quad (7)$$

The function $\log \Phi$ is concave. Indeed, with $r = \varphi/\Phi$, its second derivative is $-r(z)\{z + r(z)\} < 0$; when $z < 0$, the elementary Gaussian tail inequality gives $r(z) > -z$. Equation (7) is therefore log-concave in θ . Its minimum on each of the two half intervals is at least the smaller of its endpoint values.

For $t \in [1/25, 1/5]$, the normal arguments at $\theta = 0$ and $\theta = -1/5$ are bounded below by -2 and $-14/5$, respectively. The latter follows by maximizing $4\sqrt{t} + (2/5)/\sqrt{t}$ on that compact interval; its maximum is $14/5$. The rational inequalities

$$\begin{aligned} 24 < e^{16/5} < 25, \quad \Phi(-2) > 1/50, \quad \Phi(-14/5) > 1/500, \\ \Phi(-4/5) > 1/5, \quad \Phi(6/5) > 22/25. \end{aligned} \quad (8)$$

imply $S'_\theta(t) \geq 8/25$ on the entire negative drift half interval.

For $t \in [1/4, 1]$, the endpoint arguments at $\theta = 0$ and $\theta = 1/5$ are at least $-4/5$ and $6/5$. Hence $S'_\theta(t) \geq 352/625$ on the positive drift half interval. Since $e^{-.04t} \geq 24/25$, the two lower bounds for the derivative of the lower-killed occupation expression are

$$2 \frac{4}{25} \frac{24}{25} \frac{8}{25} - \frac{4}{125} = .066304, \quad (9)$$

$$2 \frac{3}{4} \frac{24}{25} \frac{352}{625} - \frac{54}{125} = .379008. \quad (10)$$

No derivative sample grid appears in this argument. The intervals cover every admissible constant drift, including the shared point zero.

4.3. Distant-boundary error and rational checks

The upper boundary is distant but not discarded. Under all admissible drifts, a path hitting it must have a Brownian excursion of at least $(b - u_0 - 1/5)/\sigma$. Reflection and a Gaussian tail bound give

$$p_b \leq 2 \exp \left[-\frac{(b - u_0 - 1/5)^2}{2\sigma^2} \right] = 2e^{-380.88} < 2^{-379}. \quad (11)$$

For a path-dependent difference supported on that event, the drift likelihood score has second moment at most $1/\sigma^2$. Cauchy–Schwarz bounds the derivative discrepancy between the actual doubly killed occupation integral and the extended lower-killed one by

$$\frac{20}{\sigma} \sqrt{p_b} + 2p_b < 400 2^{-189} + 2 2^{-379} < 1/1000. \quad (12)$$

Subtracting $1/1000$ from the preceding half-line bounds gives precisely $8163/125000$ and $47251/125000$. A strictly increasing continuous function on the compact drift interval has the unique maximizer $1/5$.

The file `replication/boundary.py` proves the elementary constants in (8) with exact fractions. Exponential bounds use Taylor sums and an explicit geometric tail. A Machin arctangent identity with alternating-series bounds proves a rational bound for π , and $\sqrt{2\pi} < 251/100$ follows. Gaussian lower bounds use monotone right sums of mesh $1/100$ on finite intervals, with every exponential term bounded by a rational Taylor calculation. Truncating the infinite tail only lowers a tail integral. For $\Phi(6/5)$ the integral is from zero to $6/5$, added to $1/2$. Terms are rounded downward to a common dyadic denominator before summation, with the direction of the bound preserved. The code records the resulting exact rational lower bounds and requires them to exceed all displayed thresholds.

The payoff enclosure at $\theta = 1/5$ is inherited from the unchanged validated R30 killed-kernel oracle. Its source path, Git blob hash, and the fact that it is not recomputed are recorded in `results/boundary.json`. The current theorem changes its interpretation from an endpoint evaluation to the optimum of the entire constant class. It does not assert feedback optimality or satisfy the original all-domain current-state target.

5. FULL COHORT AND AUDIT RESULTS

TABLE I: All registered class-based certificates

T	Proposal	ε	Raw regret	Certified bound	Cost saving	Bytes
4	Defer	0.01	1.61894	0.00269551	0	3897
4	Defer	0.05	1.61894	0.0134775	0	3887
4	Stress	0.01	1.61894	0.00269551	0.00227628	4660
4	Stress	0.05	1.61894	0.0134775	0.0113815	4659
4	Spline 33	0.01	4.30982e-16	4.09353e-08	0	3505
4	Spline 33	0.05	4.30982e-16	4.09353e-08	0	3504
4	Spline 129	0.01	4.14599e-16	4.09353e-08	0	3494
4	Spline 129	0.05	4.14599e-16	4.09353e-08	0	3493
4	NN 31001	0.01	0.124593	0.00269551	0	4434
4	NN 31001	0.05	0.124593	0.0134775	0	4425
4	NN 31002	0.01	0.14356	0.00269551	0	4311
4	NN 31002	0.05	0.14356	0.0134775	0	4302
4	NN 31003	0.01	0.143355	0.00269551	0	4440
4	NN 31003	0.05	0.143355	0.0134775	0	4431
8	Defer	0.01	3.35621	0.00148562	0	19038
8	Defer	0.05	3.35621	0.00804234	0	18773
8	Stress	0.01	3.35621	0.00148561	0.0174693	41810

T	Proposal	ε	Raw regret	Certified bound	Cost saving	Bytes
8	Stress	0.05	3.35621	0.00742774	0.0891575	41606
8	Spline 33	0.01	0.0057581	0.00148562	0	15821
8	Spline 33	0.05	0.0057581	0.00575812	0	12143
8	Spline 129	0.01	0.000411466	0.000411565	0	12451
8	Spline 129	0.05	0.000411466	0.000775484	0	12179
8	NN 31001	0.01	0.234831	0.00148562	0	18524
8	NN 31001	0.05	0.234831	0.00804234	0	18250
8	NN 31002	0.01	0.296282	0.00148562	0	18488
8	NN 31002	0.05	0.296282	0.00804234	0	17477
8	NN 31003	0.01	0.224246	0.00148562	0	19178
8	NN 31003	0.05	0.224246	0.00742775	0	18500
12	Defer	0.01	4.75581	0.00116122	0	195094
12	Defer	0.05	4.75581	0.00603201	0	194062
12	Stress	0.01	4.75581	0.00116031	0.0237425	370046
12	Stress	0.05	4.75581	0.00608421	0.119917	359043
12	Spline 33	0.01	0.0057581	0.00116122	0	156289
12	Spline 33	0.05	0.0057581	0.00543908	0	116384
12	Spline 129	0.01	0.000411466	0.000411565	0	91427
12	Spline 129	0.05	0.000411466	0.00105788	0	86102
12	NN 31001	0.01	0.64517	0.00116402	0	184623
12	NN 31001	0.05	0.64517	0.00643136	0	179921
12	NN 31002	0.01	0.611512	0.00115741	0	193825
12	NN 31002	0.05	0.611512	0.00616493	0	173948
12	NN 31003	0.01	0.224246	0.00116122	0	206325
12	NN 31003	0.05	0.224246	0.00603201	0	209874

The tables display rounded summaries only. Every primary certificate stores rational knots, values, policies, Bellman witnesses, and intervention-cost functions. A separate file records the exact interval-wide error and cost residual. The checker does not invoke the envelope optimizer. Its only shared components are the representation, the primitive model, and elementary exact affine operations. This is implementation separation, not a proof assistant or an assertion of independence from all shared arithmetic code.

Eight deliberately malformed inputs are rejected: changed model, absent model, negative tolerance, missing date, missing interval, changed endpoint, false cost, and infeasible action. Tests also retain point spikes and affine preimages and compare the exact two-period Bellman solution with exhaustive adaptive scenario-tree enumeration at four rational restart states. The test count is not substituted for the whole-state inequalities in the certificates.

6. HISTORICAL PRESERVATION AND CURRENT CLAIMS

The document `HISTORY_R32.pdf` concatenates the five R30 submission objects without altering their pages: article, supplement, response, computational record, and the earlier historical annex. The repository retains their original source and result paths. Thus the previous action-independent operating-mode study, inventory comparison, Adam transport and rollback derivations, central-state search, active-boundary quadrature failures, and earlier referee responses are not deleted to simplify the current narrative.

In particular, the unfavorable exhaustive inventory timing ratios, the 26/60 raw passes in the prior operating-mode cohort, and the exact symbolic and online-guard controls remain available. They are historical evidence, not results of the new action-dependent model. The original whole-domain bound remains 7.181834580823298 and its target remains .01. No new result is relabeled as a 47-coordinate derivative bridge, a high-dimensional certificate, or neural-specific acceleration. The constructive coupled certificates and the global active-boundary scalar theorem are additional results with explicitly delimited objects.

7. UNRESTRICTED SUPPORT CERTIFICATES AND THE GLOBAL COST GAP

This appendix supplements Section 5 of the article. The original upper witness and all new support objects use the same rational representation: affine formulas on open cells and independently stored values at their knots. Jumps induced by a compiled installed rule are allowed; measure-zero values are not discarded from any restart guarantee.

7.1. Verification instructions and proof dependencies

A support certificate stores the exact model, its multiplier, U , the selector γ , the defect bounds d_t , the installed policy, its support-optimal deployment, W , and that deployment's operating and implementation values. The checker first verifies the model and horizon, legal point and interval actions, and the terminal conditions. It checks $T^a U_{t+1} \leq U_t$ for every action and the displayed defect inequality for γ . The propagated errors and $L = U - b$ are then reconstructed by the checker. No file purporting to contain V is read.

For every support object, the checker verifies that every action-continuation alternative in the support Bellman equation lies below W_t , and that the stored deployment attains equality. It also checks the own-policy Bellman equalities for J, C and the identity $W = \lambda J - C$. These checks imply unrestricted support optimality by backward induction. They do not require a maximizing-envelope computation. The same check proves that no adaptive or randomized action choice can exceed W : conditional expectation and a convex combination preserve the upper comparison.

An assembled certificate stores a feasible deployment, its J, C , its feasibility upper witness, the lower-cost function B , and a selector identifying the attaining lower-bound term on every open cell and knot. It binds the complete ordered multiplier portfolio by SHA-256 hashes of canonical uncompressed JSON. Every support certificate must first pass its own check; hashes alone do not establish mathematical validity. The assembled checker verifies the lower envelope by comparing each candidate term to B and checking equality with the selected term. It verifies $B \leq C$, every restart's operating tolerance, and the exact integrated gap. A separate check compares the selected J to the installed own-policy value to establish operating preservation when it holds.

The implementation of the lower selector accepts integer indices for lower-bound terms, not action labels. Its feasible policy selector accepts only the three legal maintenance actions. Confusing these objects could produce a silently spliced policy, so their validations are distinct. The terminal cost is zero, the terminal operating value is g , and the terminal support value is λg .

7.2. Feasibility, finite selection, and the scope of the lower bound

The sufficient-multiplier proposition guarantees that the declared price 4096 has a feasible support selector for every installed rule in the declared model and horizon set. This is an ex ante statement from the bounded implementation cost and witness defect, not an assertion inferred

from the 42 observed passes. The independent audit verifies the rational sufficient bound for each horizon. Every actual serialized candidate is nevertheless tested directly at all dates and states.

The selected primal policy minimizes integrated implementation cost only among the finite set of individually verified complete policies, including the class baseline. This selection is not an optimization over all feasible policies. In contrast, every lower-bound term applies to every feasible policy, so the assembled cost interval bounds the unrestricted constrained optimum. The two scopes must not be exchanged. In particular, the computation does not assume that a support-optimal policy exists for every constrained-optimal operating budget, or that a single common multiplier is a complete dual for the simultaneous continuum of restart constraints.

The pointwise maximum of lower bounds remains a lower bound even when no single policy attains it everywhere. Integration is valid because every admissible policy dominates that pointwise function. Equality of the integrated bounds proves optimality for the stated initial distribution; without a pointwise zero gap it does not by itself prove that the deployment minimizes cost at each individual restart. The supplied all-restart cost-gap envelope is an additional reported quantity, not a consequence of a small integrated gap.

All implementation costs are nonnegative. At zero cost, the integrated gap vanishes for the declared spline cases. Their stronger every-restart zero-cost result follows directly because the retained incumbent has identically zero intervention, not merely zero cost almost everywhere. Isolated action differences in other deployments remain part of every feasibility check even when they do not affect a uniform-distribution integral.

7.3. Costs, unfavorable cases, and preservation of attempts

The eight multiplier problems are all solved and charged, including every candidate that fails an operating tolerance. The common witness is produced from primitives; no precomputed optimal-value file is supplied. The classical class candidate is rebuilt and checked on the same machine using the complete-cost constructor. Every proposed support policy is independently evaluated rather than assigned a cost from its optimization label. The final byte-level audit is an additional cost, reported separately from the checks within construction. Exact operating dynamic programming is run only as a recorded comparator and diagnostic.

The replication stores all 168 support outcomes and all 42 final intervals, not only the neural improvements. Timing records distinguish fitting of the previously installed proposals from revising those fixed proposals now. They do not add incompatible machine timings to obtain a purported speedup. The frozen R32 records and staged R33 source archive remain losslessly preserved. R34 execution records are new and separately identified; no unavailable prior run is reconstructed or asserted as evidence. Compressed encodings and their checksums remain explicitly distinguished from canonical uncompressed certificate bytes.

The checker rejects eight malformed support certificates and eight malformed assembled certificates, including changed endpoint values, missing dates and intervals, false costs or lower bounds, illegal actions, bad multiplier signs, incorrect terminal conditions, and altered support hashes. Four two-period rational restart states are independently compared with all 27 contingent two-step action trees. The final read-only audit disables the optimizing envelope routine, so a hidden repeat of the constructor's optimization is an error rather than a fallback.

7.4. Complete unrestricted-cost cohort

TABLE II: Complete globally bounded portfolio; all 42 declared configurations

T	ε	Installed	λ	Cost lower	Cost upper	Gap	Saved
4	0.01	Defer	64	4.312943	4.937421	0.624478	0.000051
4	0.05	Defer	Class	3.664041	4.936309	1.272268	0.000000
4	0.01	Stress	256	3.628917	4.209701	0.580784	0.000202
4	0.05	Stress	Class	3.000666	4.205319	1.204652	0.000000
4	0.01	Spline 33	0	0.000000	0.000000	0.000000	0.000000
4	0.05	Spline 33	0	0.000000	0.000000	0.000000	0.000000
4	0.01	Spline 129	0	0.000000	0.000000	0.000000	0.000000
4	0.05	Spline 129	0	0.000000	0.000000	0.000000	0.000000
4	0.01	NN 31001	256	0.074641	0.114005	0.039364	0.003946
4	0.05	NN 31001	64	0.025368	0.094890	0.069522	0.013065
4	0.01	NN 31002	256	0.109583	0.152287	0.042704	0.003946
4	0.05	NN 31002	64	0.045915	0.133172	0.087257	0.013065
4	0.01	NN 31003	256	0.083932	0.123969	0.040037	0.003946
4	0.05	NN 31003	64	0.030930	0.104854	0.073924	0.013065
8	0.01	Defer	64	10.203910	10.823575	0.619666	0.001457
8	0.05	Defer	Class	9.445140	10.815918	1.370779	0.000000
8	0.01	Stress	256	8.887172	9.535807	0.648635	0.005663
8	0.05	Stress	64	7.942794	9.491694	1.548900	0.014597
8	0.01	Spline 33	0	0.000000	0.000000	0.000000	0.029666
8	0.05	Spline 33	0	0.000000	0.000000	0.000000	0.000000
8	0.01	Spline 129	0	0.000000	0.000000	0.000000	0.000000
8	0.05	Spline 129	0	0.000000	0.000000	0.000000	0.000000
8	0.01	NN 31001	256	0.205701	0.533681	0.327980	0.030524
8	0.05	NN 31001	64	0.060208	0.415981	0.355774	0.087152
8	0.01	NN 31002	256	0.095675	0.397532	0.301857	0.044178
8	0.05	NN 31002	64	0.000360	0.282891	0.282531	0.086746
8	0.01	NN 31003	1024	0.352314	1.017217	0.664903	0.014023
8	0.05	NN 31003	64	0.045344	0.477618	0.432274	0.442321
12	0.01	Defer	64	15.001167	15.637251	0.636084	0.001873
12	0.05	Defer	Class	14.118948	15.627451	1.508503	0.000000
12	0.01	Stress	256	13.140504	13.797365	0.656861	0.009866
12	0.05	Stress	64	11.957376	13.741126	1.783750	0.016973
12	0.01	Spline 33	0	0.000000	0.000000	0.000000	0.029366
12	0.05	Spline 33	0	0.000000	0.000000	0.000000	0.000880
12	0.01	Spline 129	0	0.000000	0.000000	0.000000	0.000000
12	0.05	Spline 129	0	0.000000	0.000000	0.000000	0.000000
12	0.01	NN 31001	1024	2.620836	3.894877	1.274042	0.065588
12	0.05	NN 31001	256	1.524265	3.520192	1.995927	0.214315
12	0.01	NN 31002	256	1.880895	2.651884	0.770989	0.272947
12	0.05	NN 31002	64	1.033237	2.262773	1.229536	0.452344
12	0.01	NN 31003	1024	0.480953	1.514982	1.034029	0.008744
12	0.05	NN 31003	64	0.041222	0.674484	0.633262	0.730212

Displayed decimals are descriptive rounded summaries of exact rational bounds. The machine-readable fractions, not outward rounding of these printed decimals, define the certificates. “Class denotes the freshly reconstructed local-class candidate. Saved cost is relative to that class, not a claim that a positive-cost deployment is globally optimal. The 12 spline cases have identically zero cost and exact zero global gap. All selected deployments satisfy their original all-restart operating tolerance and separately preserve installed operating values.

8. RESTART-PROPAGATED LOWER BOUNDS: VERIFIER AND COMPLETE COHORT

8.1. *Precise feasibility class*

For a Markov randomized rule, the conditional expected continuation values $J_t^\pi(x)$ and $C_t^\pi(x)$ include subsequent action randomization as well as the model shocks. Feasibility requires $J_t^\pi(x) \geq V_t(x) - \varepsilon$ for every t, x . It does not require each action in the support of the rule to satisfy this inequality on its own. For a history-dependent extension, impose the analogous conditional expected inequality after every history for which a continuation is being certified. An initial-distribution-only expected constraint defines a different problem and does not justify recursive use of future feasible-continuation lower bounds. These definitions explain exactly the class covered by Theorem 4.

8.2. *Verifiable finite object*

The serialized closure stores the model and version, horizon, tolerance, installed-rule name, the canonical input-certificate SHA-256, the complete ordered support-hash list, local multipliers, every A_t , every m_t^μ , an action selector for each minimum, and the source selector for the outer maximum. The terminal A_T is identically zero. Stored rational knots include zero and one, are strictly ordered, and have separate function values. Every open interval has an affine formula. This representation distinguishes values at ties from their one-sided limits and remains valid when an installed rule induces jumps.

The fresh-process checker first verifies all support and base certificates, rather than treating their hashes as proofs of validity. A hash identifies the exact object; Bellman comparisons establish its validity. It then reconstructs the action expressions in equations (22)–(23). For each local minimum it checks domination by every action expression and equality at its supplied selector. For the outer maximum it checks domination of every source term and equality at the source selector. It verifies $B \leq A \leq C^\pi$, the terminal condition, horizon and multiplier identities, and both reported integrals. All selector values must be legal integers, including those at isolated knots. Every comparison examines the union of rational knots and both limits of each resulting open affine interval. The optimizing envelope routine is disabled in the constructor module, the shared arithmetic module, and the base-audit module.

These comparisons are sufficient for exact min/max identities: a supplied minimum lying below all alternatives and equaling one alternative is the true minimum, and the dual argument proves a maximum. No optimizer is needed to check these certificates. The checker still uses exact partition overlays, rational arithmetic, and file parsing; these costs are timed. It is an independently executed checker sharing an exact arithmetic library, not a formally verified proof assistant or a fully independent second arithmetic implementation.

8.3. *Adversarial and finite-tree tests*

The mutation suite requires rejection of a negative local multiplier, an omitted multiplier row, an inflated isolated lower-bound point, an inflated isolated minimum point, an illegal minimizing action, a nonintegral outer selector, an incorrect terminal value, a missing date, a changed base hash, a changed support hash, a false integrated gap, and a changed tolerance. Eight support mutations and eight base-summary mutations are also retained. A malformed object is never repaired silently by the checker.

At each of four rational initial states the finite-tree test enumerates the three first actions and all nine shock-contingent second-action pairs. It next enumerates all 378 unordered equal mixtures of the 27 plans. Mixtures sharing the first action use the conditional average second-stage

value after each observed shock; different first actions reveal which branch of the mixture was chosen, and each corresponding continuation is checked separately. This realizes a behavioral randomized rule with no additional remembered private-state constraint. The test rejects no action merely for failing an individual operating test. Only plans or mixtures satisfying the actual conditional operating inequalities enter the lower-bound comparison. The stored test record reports both the total and the eligible counts, so nonvacuity is auditable.

8.4. Complete exact outcomes

TABLE III: All restart-propagated cost intervals

T	ϵ	Installed	Old lower	New lower	Upper	New gap	Gain
4	0.01	Defer	4.312942	4.312942	4.937421	0.624479	0.000000
4	0.05	Defer	3.664040	3.664040	4.936310	1.272269	0.000000
4	0.01	Stress	3.628916	3.636007	4.209702	0.573694	0.007091
4	0.05	Stress	3.000666	3.017302	4.205319	1.188016	0.016636
4	0.01	Spline 33	0.000000	0.000000	0.000000	0.000000	0.000000
4	0.05	Spline 33	0.000000	0.000000	0.000000	0.000000	0.000000
4	0.01	Spline 129	0.000000	0.000000	0.000000	0.000000	0.000000
4	0.05	Spline 129	0.000000	0.000000	0.000000	0.000000	0.000000
4	0.01	NN 31001	0.074641	0.076202	0.114005	0.037803	0.001560
4	0.05	NN 31001	0.025367	0.025367	0.094890	0.069523	0.000000
4	0.01	NN 31002	0.109582	0.111143	0.152287	0.041143	0.001560
4	0.05	NN 31002	0.045915	0.045915	0.133172	0.087257	0.000000
4	0.01	NN 31003	0.083931	0.085492	0.123970	0.038477	0.001560
4	0.05	NN 31003	0.030930	0.030930	0.104855	0.073925	0.000000
8	0.01	Defer	10.203909	10.203924	10.823576	0.619652	0.000014
8	0.05	Defer	9.445139	9.445139	10.815919	1.370779	0.000000
8	0.01	Stress	8.887171	8.893666	9.535807	0.642140	0.006494
8	0.05	Stress	7.942793	7.954458	9.491694	1.537236	0.011664
8	0.01	Spline 33	0.000000	0.000000	0.000000	0.000000	0.000000
8	0.05	Spline 33	0.000000	0.000000	0.000000	0.000000	0.000000
8	0.01	Spline 129	0.000000	0.000000	0.000000	0.000000	0.000000
8	0.05	Spline 129	0.000000	0.000000	0.000000	0.000000	0.000000
8	0.01	NN 31001	0.205700	0.324516	0.533682	0.209165	0.118815
8	0.05	NN 31001	0.060207	0.102483	0.415982	0.313498	0.042276
8	0.01	NN 31002	0.095675	0.211931	0.397533	0.185601	0.116256
8	0.05	NN 31002	0.000359	0.040296	0.282892	0.242595	0.039936
8	0.01	NN 31003	0.352313	0.483252	1.017217	0.533964	0.130939
8	0.05	NN 31003	0.045343	0.068233	0.477618	0.409385	0.022889
12	0.01	Defer	15.001166	15.001166	15.637251	0.636084	0.000000
12	0.05	Defer	14.118947	14.118947	15.627451	1.508503	0.000000
12	0.01	Stress	13.140504	13.149369	13.797366	0.647996	0.008865
12	0.05	Stress	11.957375	11.967621	13.741127	1.773505	0.010245
12	0.01	Spline 33	0.000000	0.000000	0.000000	0.000000	0.000000
12	0.05	Spline 33	0.000000	0.000000	0.000000	0.000000	0.000000
12	0.01	Spline 129	0.000000	0.000000	0.000000	0.000000	0.000000
12	0.05	Spline 129	0.000000	0.000000	0.000000	0.000000	0.000000
12	0.01	NN 31001	2.620835	2.711965	3.894878	1.182912	0.091130
12	0.05	NN 31001	1.524265	1.566705	3.520193	1.953488	0.042439
12	0.01	NN 31002	1.880894	2.020437	2.651884	0.631446	0.139543
12	0.05	NN 31002	1.033237	1.137697	2.262773	1.125075	0.104460
12	0.01	NN 31003	0.480952	0.742636	1.514983	0.772346	0.261684
12	0.05	NN 31003	0.041222	0.077511	0.674484	0.596973	0.036289

Lower bounds and gains are rounded downward; upper costs and excess-cost bounds are rounded upward. Exact fractions and all knot values are retained in the machine-readable certificates. The upper deployment is unchanged. A printed zero gain can conceal a positive gain smaller than 10^{-6} ; exact positivity is recorded in the summary.

The lower-bound construction does not change the already verified deployed policy. Its source selectors are proof data, not an action policy obtained by splicing multipliers. This is why the new global cost interval can be compared directly with the previous interval while keeping the upper policy, its operating tolerance, and its occupancy law fixed. All recorded nonzero gaps remain explicit. Raw binary64 neural execution and the full stopped-economy feedback certificate are separate mathematical objects and are not covered by these exact maintenance-model checks.